TAMIL NADU ELECTIONS 2026

Full Data Science Project

Project Brief, Data Overview, Findings & Roadmap

A team project using real election data from Tamil Nadu 2026 Assembly Elections

234 Constituencies | 57.3 Lakh Voters | 85.1% Turnout | 4,023 Candidates

Polling: April 23, 2026 | Results: May 4, 2026

Team of 5 Members | May 2026

1. What Is This Project?

This is a full data science project built entirely on real data from the Tamil Nadu 2026 Legislative Assembly Elections. The election was held on April 23, 2026, results were declared on May 4, 2026, and we are building this project right now — using actual data that was collected directly from the Election Commission of India and the Tamil Nadu Chief Electoral Officer.

The goal is not just to make a dashboard. The goal is to do what a real data team does in the industry:

- A Data Engineer builds the data pipeline — cleaning, merging, and storing raw data properly
- A Data Analyst finds patterns, creates charts, and explains what happened
- A Data Scientist builds statistical models and predicts things — like which seats were most likely to flip
- An AI Engineer adds intelligent features — like a chatbot that answers questions from the data

By doing all four roles in one project, every team member gets exposure to the full data career pathway. When you go for interviews, you can show this one project and demonstrate skills in all four job roles.

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2. Why This Project Matters

2.1 For Society

Tamil Nadu elections affect 7.7 crore people. But most people do not understand what the data actually says. This project makes election data accessible, understandable, and useful for:

- Citizens who want to understand how their constituency voted
- Journalists who need data-backed stories about the election
- Researchers studying voter behaviour, gender participation, and political patterns
- Policymakers who want to understand which regions have low turnout and why
- Young voters who are curious about how democracy works in numbers

The 2026 Tamil Nadu election had the highest voter turnout ever recorded — 85.1%. A new party (TVK, led by actor Vijay) entered elections for the first time. Female voter participation was at a historic high. These are significant events and our project quantifies exactly what happened, where, and why.

2.2 For Your Career

Most fresher portfolios show the Titanic dataset or IPL scores. Those are practice datasets that every interviewer has seen thousands of times. This project is different:

- Real data from an event that happened 3 weeks ago
- Your own collected datasets combined with public official data
- Analysis that has never been done before at this level of detail
- Covers all four job roles: Data Analyst, Data Engineer, Data Scientist, AI Engineer
- Produces actual deliverables: dashboard, report, GitHub repo, ML models, chatbot

When a recruiter asks 'show me a project you have done', this is the answer that stands out.

3. Data We Have — Files and What They Contain

We currently have 5 datasets uploaded and verified. Here is exactly what each file contains and what we will use it for.

File 1 — eci_results_tamilnadu_2026.csv

Source: Election Commission of India official results portal. Date: May 4, 2026.

Column	What It Means	Used For
Code	Unique constituency code (e.g. S221)	Joining with other files
Constituency	Name of the constituency	All analyses
Candidate	Full name of each candidate	Candidate analysis
Party	Party name (105 unique parties)	Alliance analysis
EVM Votes	Votes received from Electronic Voting Machine	Vote count analysis
Postal Votes	Votes received by postal ballot	Voter behaviour
Total Votes	EVM + Postal combined	Win/loss, margins
% Votes	Percentage of total votes this candidate got	Swing analysis vs 2021
Round	Counting round number	Counting pattern analysis
Last Updated Date	04/05/2026 for all rows	Data verification

Total rows: 4,257 (all candidates across 234 constituencies). Zero missing values. This is our primary results file.

File 2 — voters.xlsx

Source: Tamil Nadu Chief Electoral Officer. Final Electoral Roll as of February 23, 2026.

Column	What It Means	Used For
District No.	District number (1 to 38)	District grouping
District Name	Name of the district	Regional analysis
AC No.	Assembly Constituency number (1 to 234)	Primary join key

Column	What It Means	Used For
Name of Assembly Constituency	Official constituency name	Merging files
Male	Total male electors registered	Gender analysis
Female	Total female electors registered	Gender analysis
Third Gender	Total transgender electors registered	Trans voter analysis
Total Electors	Sum of all registered voters	Turnout calculation
Total_Candidates	How many candidates contested	Competition analysis

File 3 — 2026_voters_voted.xlsx

Source: Tamil Nadu Chief Electoral Officer. This is your friend's dataset — actual voting counts for the election day.

Column	What It Means	Used For
AC No.	Assembly Constituency number	Primary join key
AC Name	Constituency name	Verification
Total Electors (Male/Female/Trans/Total)	Registered voters on election day	Turnout denominator
Electors who Voted (Male/Female/Trans)	How many actually voted by gender	Gender turnout analysis
POSTAL	Postal votes cast	Postal vote trends
TOTAL (voted)	All votes cast including postal	Official turnout number
POLL %	Official turnout percentage	Key metric — 85.1% avg
NOTA Votes	None of the Above votes	Voter dissatisfaction index
Valid Votes Polled	Votes that were counted as valid	Actual result base

File 4 — 2021_election_reults.xlsx

Source: Election Commission of India archive. Results from the 2021 Tamil Nadu Assembly Election for all 234 constituencies.

Column	What It Means	Used For
State/UT Name	Tamil Nadu	Verification
AC No.	Constituency number	Primary join key
AC Name	Constituency name	Joining with 2026 data
Candidate Name	All 2021 candidates	Finding seat flips
Sex	Candidate gender (Male/Female)	Gender representation analysis
Age	Candidate age	Candidate profile analysis
Category	General/SC/ST	Reservation analysis
Party	Party they represented in 2021	Alliance comparison 2021 vs 2026
General	EVM votes in 2021	Vote share swing calculation
Postal	Postal votes in 2021	Comparison
Total	Total votes in 2021	Seat flip identification
% Votes Polled	Vote share percentage in 2021	Swing analysis
Total Electors	Registered voters in 2021	Turnout growth calculation

File 5 — candidates_list__-_2026.csv

Source: Official candidate nomination data from CEO Tamil Nadu. This is the master candidate list with party and alliance mapping.

Column	What It Means	Used For
AC_No	Constituency number	Primary join key
AC_Name	Constituency name	Verification
District	District the constituency is in	Regional grouping
Reserved	GEN / SC / ST reservation status	Reservation analysis
Sl_No	Candidate serial number on ballot	Ballot position analysis
Candidate_Name	Full candidate name	Joining with results
Party_Full	Full party name	Party identification
Party_Short	Short party code (DMK, AIADMK, TVK etc.)	Charts and labels
Alliance	Which alliance they belong to (INDIA/SPA, NDA, Solo)	Key for alliance analysis

Column	What It Means	Used For
Symbol	Election symbol (Rising Sun, Two Leaves etc.)	Voter reference

4. Data Still Needed

We have 5 of the 10 planned files. Here are the remaining files and where to get them.

File Name	What It Contains	Where to Get It	Priority
constituency_master.csv	Region (North/South/West/Central/Chennai), urban/rural tag for all 234 seats	Build manually using district list — 1 hour work	HIGH — needed for maps
party_alliance_master.csv	Clean mapping of all 105 parties to their alliance	Already partially in candidates file — needs deduplication	HIGH — needed for DMK+ vs AIADMK+ charts
cabinet_2026.csv	New ministers, portfolios, constituencies they won from	The Hindu / Dinamalar cabinet swearing-in articles	MEDIUM — for cabinet analysis
district_socioeconomic.csv	Literacy rate, HDI, per capita income per district	NITI Aayog district HDI report / Census 2011	LOW — only needed in Week 3 for ML
candidates_detail.csv	Age, gender, criminal cases, assets of candidates	myneta.info — TN 2026 section	LOW — optional interesting finding

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5. What We Will Discover — The 20 Key Findings

These are the specific insights our analysis will produce. Each finding is tied to actual data we have. These are not guesses — they are questions that the data can answer.

Group A — Voter Participation Findings (from Files 1, 2, 3)

- **Finding 1 — Constituency-wise turnout ranking**
    - All 234 constituencies ranked by turnout %. Top 10 and bottom 10 identified.
    - We already have the exact POLL% column in the 2026_voters_voted file.
    - Example output: 'Gudalur had 92.4% turnout — highest in TN 2026'

- **Finding 2 — Gender gap in voter registration**
    - In how many of 234 constituencies do female registered voters outnumber male?
    - Which district has the highest female voter share?
    - From voters.xlsx — straightforward calculation.
- **Finding 3 — Gender gap in actual voting**
    - Did female voters turn out at higher rates than male voters on election day?
    - Constituency-wise: Male voted / Male registered vs Female voted / Female registered
    - From 2026_voters_voted.xlsx — gender-split turnout available.
- **Finding 4 — Transgender voter participation map**
    - First-ever constituency-level map of transgender voter registration in TN 2026
    - Which constituency had the most transgender voters? Which had zero?
    - This is a socially significant finding — never been visualised for Tamil Nadu at this level
- **Finding 5 — NOTA votes analysis**
    - Which constituencies had the highest NOTA votes?
    - High NOTA = voter dissatisfaction with all candidates. This is politically meaningful.
    - NOTA column directly available in 2026_voters_voted.xlsx
- **Finding 6 — Turnout growth from 2021 to 2026**
    - 2021 turnout per constituency available in 2021 results file (Total / Total Electors)
    - Compare with 2026 POLL% to find: which seats grew turnout the most? Which declined?

Group B — Election Results Findings (from Files 1, 5)

- **Finding 7 — Alliance-wise seat count**
    - DMK alliance (INDIA/SPA) vs NDA vs TVK vs Solo parties — total seats won
    - Candidates file has Alliance column — combine with results to get winner alliance
    - This is the most-watched number in TN politics
- **Finding 8 — TVK debut performance analysis**
    - Actor Vijay's Tamilaga Vettri Kazhagam (TVK) contested all 234 seats for the first time
    - How many did they win? In which districts? What was their average vote share?
    - Did they take votes from DMK or AIADMK — or from new voters?
    - This is the single biggest political story of 2026 TN elections
- **Finding 9 — Winning margins distribution**
    - How many seats were won by under 2,000 votes? Under 5,000? Under 10,000?

- What was the largest winning margin? The smallest?
- A histogram of all 234 margins — tells the story of how competitive this election was

- **Finding 10 — Seats that flipped from 2021 to 2026**
  - Compare winners in 2021 vs 2026 for all 234 seats
  - Which seats changed party? Which changed alliance? How many total flips?
  - Which direction did they flip — DMK to TVK? AIADMK to DMK?

- **Finding 11 — Geographic strongholds map**
  - TN map coloured by winning alliance per constituency
  - Expected: DMK strong in north TN and Chennai, AIADMK in western districts
  - Did these patterns hold in 2026, or did TVK disrupt them?

- **Finding 12 — Reserved constituency performance**
  - SC reserved seats vs GEN seats: did turnout, margins, or party performance differ?
  - Reserved column available in candidates file

Group C — Statistical Findings (Data Scientist — Week 3)

- **Finding 13 — Does female voter share predict higher turnout? (Correlation)**
  - Scatter plot: x = female voter registration %, y = actual turnout %
  - Calculate Pearson correlation coefficient. If strong positive: publish this finding.

- **Finding 14 — Turnout prediction model (Regression)**
  - Inputs: district, gender ratio, urban/rural, number of candidates, HDI (week 3)
  - Output: predicted turnout %
  - Key finding: 'These 3 factors explain 70% of turnout variation across TN'

- **Finding 15 — Constituency clusters (K-Means)**
  - Group all 234 constituencies into 3-4 behaviour types based on voting patterns
  - Name them: 'Urban high-turnout seats', 'Western AIADMK base', 'New voter TVK seats'
  - This is a genuine research contribution — no one has done this for 2026

- **Finding 16 — Seat flip predictor (Classification model)**
  - Train a model: did a seat flip from 2021 to 2026?
  - Inputs: 2021 margin, turnout change, TVK presence, gender ratio
  - Report AUC score — even 0.68 is meaningful for political data

- **Finding 17 — Postal vote pattern analysis**
  - Which constituencies had unusually high postal votes?

- Did postal votes change the winner in any close race?
- Postal votes column available in both 2021 and 2026 results

Group D — Special Interest Findings

- **Finding 18 — Number of women candidates vs women winners**
  - From the candidates and results files: how many women contested? How many won?
  - Which party fielded the most women candidates? Which had best win rate?
- **Finding 19 — Candidates-per-constituency and its effect on margins**
  - Did constituencies with more candidates have smaller winning margins?
  - Total_Candidates column in voters.xlsx
- **Finding 20 — Vote splitting analysis**
  - In seats where TVK was present, did the winner have a lower vote share than in 2021?
  - Did TVK split the opposition vote and help DMK, or did they split the ruling party vote?
  - This is the most analytically sophisticated finding in the project

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6. Final Outputs — What We Will Deliver

Output 1 — Interactive Dashboard (Streamlit, hosted on the web)

A live web application anyone can open with a link. Built in Python using Streamlit. Filters by party, district, region, alliance.

Pages inside the dashboard:

- Home page: TN map coloured by winning party + key stats (total seats per alliance)
- Turnout explorer: filter by district, see constituency-wise turnout bar chart
- Voter demographics: male/female/trans registration and turnout breakdown
- Results deep-dive: search any constituency, see all candidates and their votes
- 2021 vs 2026 comparison: flipped seats, turnout change, vote share swing

Output 2 — Professional PDF Report

A 15-20 page analyst report. Written as if delivering to the Election Commission or a major newspaper. Contains:

- Executive summary (1 page): the 5 most important findings, plainly stated
- Methodology section: how data was collected, cleaned, and analysed
- 10+ key findings with charts embedded
- TVK debut analysis: dedicated section on the new party's performance
- Gender participation analysis: a strong social finding
- Conclusion and recommendations for future election monitoring

Output 3 — GitHub Repository

A clean, professional GitHub repository. This is what every interviewer will open first.

- Folder: /data — all raw and cleaned CSV files
- Folder: /notebooks — Jupyter notebooks for each analysis phase
- Folder: /dashboard — Streamlit dashboard code
- Folder: /models — ML model notebooks (week 3)
- README.md — explains the project, findings, how to run it
- requirements.txt — all Python packages needed

Output 4 — Jupyter Notebooks (Analysis documentation)

One Jupyter notebook per phase. Shows thinking process step by step — data loading, cleaning, analysis, charts.

- notebook_01_data_cleaning.ipynb
- notebook_02_voter_analysis.ipynb
- notebook_03_results_analysis.ipynb
- notebook_04_2021_vs_2026_comparison.ipynb
- notebook_05_ml_models.ipynb (week 3)

Output 5 — Election Q&A Chatbot (AI Engineer stretch goal)

A Streamlit app where anyone types a question about the election and gets an answer from the data. Powered by Claude API.

- User asks: 'Who won Madurai North?' — Bot answers from our results CSV
- User asks: 'Which party won most seats in Salem district?' — Bot aggregates and answers
- User asks: 'What was the turnout in TVK's best constituency?' — Bot analyses and answers

This is the AI Engineer deliverable. Very few fresher portfolios show this capability. It demonstrates RAG (Retrieval Augmented Generation) — a core AI engineering skill.

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7. The Four Job Roles — What Each Role Does in This Project

Role	What You Do in This Project	Tools Used	Skills Demonstrated
Data Engineer	Collect data, clean it, fix errors, merge files, build a database	Python, Pandas, SQLite, BeautifulSoup	ETL pipeline, data cleaning, schema design
Data Analyst	Find patterns, create charts, answer business questions, build dashboard	Pandas, Matplotlib, Seaborn, Plotly, Streamlit	EDA, data visualisation, storytelling with data
Data Scientist	Build predictive models, run statistical tests, cluster constituencies	Scikit-learn, Scipy, Statsmodels	Regression, classification, clustering, hypothesis testing
AI Engineer	Build the Q&A chatbot using election data and Claude API (LLM)	LangChain / Claude API, Streamlit, Python	RAG, prompt engineering, LLM integration

Each team member can choose to focus on one role or learn all four. The project is structured so that all four roles work together on the same dataset — just like a real company data team.

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8. Team Work Distribution

We are a team of 5. Here is a suggested way to split the work:

Team Member	Primary Role	Responsibility in This Project
Member 1 (Team Lead)	Data Engineer	Data pipeline, file cleaning, merging, database setup. Produces the clean final CSVs everyone else uses.
Member 2	Data Analyst — Voter	Voter demographic analysis: turnout, gender, trans voters, NOTA, district patterns. Builds voter section of dashboard.
Member 3	Data Analyst — Results	Election results analysis: alliance seats, TVK performance, margins, flips, maps. Builds results section of dashboard.
Member 4	Data Scientist	Statistical analysis and ML models: regression, clustering, seat flip predictor. Week 3 work.
Member 5	AI Engineer + Report	Election Q&A chatbot + writes the final PDF report. Combines all findings into the deliverable.

Everyone should participate in Phase 1 (data cleaning) together so all five understand the data fully before specialising.

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9. Four-Week Project Timeline

Week	Phase	What Gets Done	Output
Week 1	Data Engineering	Clean all 5 files, merge them using AC No. as the key, build alliance mapping, create final master CSV, store in SQLite database	Clean master dataset ready for analysis
Week 2	Data Analysis	All 20 findings from Groups A and B, all charts, TN map visualisation, 2021 vs 2026 comparison	Dashboard (voter + results pages) + Jupyter notebooks
Week 3	Data Science	Statistical models: regression, clustering, seat flip predictor, correlation tests, hypothesis testing	ML notebooks + model accuracy reports
Week 4	Packaging	Final PDF report, clean GitHub repo, deploy dashboard to Streamlit Cloud, build chatbot	All 5 deliverables complete and shareable

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10. Tools and Technologies

Tool	What It Does	Used In Phase
Python 3.11+	Main programming language for everything	All phases
Pandas	Data cleaning, merging, analysis	Phase 1 and 2
Matplotlib & Seaborn	Static charts for the report	Phase 2
Plotly	Interactive charts for the dashboard	Phase 2
Folium / Geopandas	TN constituency map visualisation	Phase 2
Streamlit	Build and host the interactive dashboard — free hosting	Phase 2 and 4
Scikit-learn	Regression, clustering, classification ML models	Phase 3
Scipy / Statsmodels	Statistical tests (t-test, Pearson correlation)	Phase 3
SQLite / PostgreSQL	Store cleaned data in a proper database	Phase 1
Jupyter Notebook	Write and document analysis step by step	Phase 2 and 3
GitHub	Version control and portfolio hosting	All phases
Claude API	Power the election Q&A chatbot	Phase 4

11. How to Present This Project in Interviews

When an interviewer asks 'tell me about a project you have done', here is what to say:

'I led a team of 5 to build a full data science project on the Tamil Nadu 2026 Assembly Elections — real data from an event that happened last month. We collected data from the Election Commission of India and CEO Tamil Nadu, covering all 234 constituencies, 57 lakh voters, and 4,023 candidates. I worked on the data engineering layer — built a pipeline to clean and merge 5 different datasets, created an alliance mapping for 105 parties, and loaded everything into a structured database. The project covers data analysis, machine learning models to predict turnout and seat flips, and an AI-powered chatbot that answers questions from the election data using a language model API.'

Then show: GitHub link, Streamlit dashboard link, PDF report.

This answer covers: real data, team collaboration, data engineering, analysis, ML, AI integration — everything a hiring manager wants to hear.

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12. Immediate Next Steps

Here is what the team should do this week, in order:

1. Share this document with all 5 team members so everyone has the same understanding
2. Assign roles to each team member (see Section 8)
3. Set up a shared GitHub repository — everyone clone it locally
4. All five members install Python, Pandas, Jupyter Notebook on their machines
5. Member 1 starts Phase 1: load all 5 CSVs into Python, check shapes and null counts
6. Collect the 3 missing files: constituency_master, cabinet_2026, district_socioeconomic
7. Weekly check-in: 30 minutes every Sunday — what did we complete, what is blocked

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